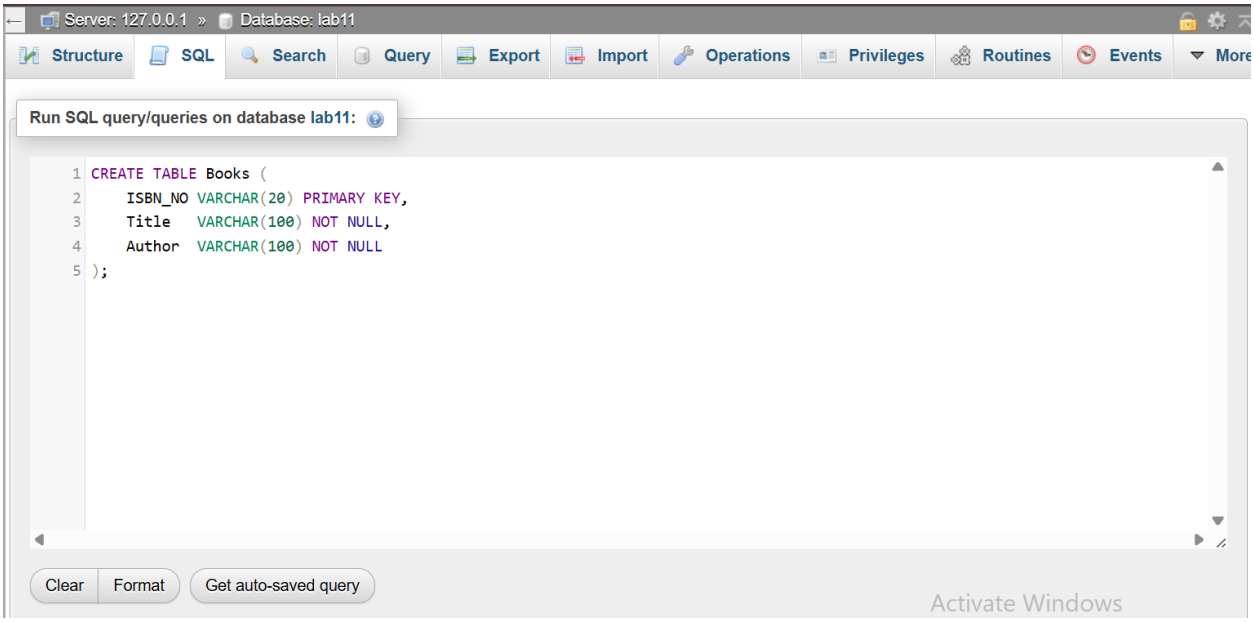
DBS LAB 11

Task 1:

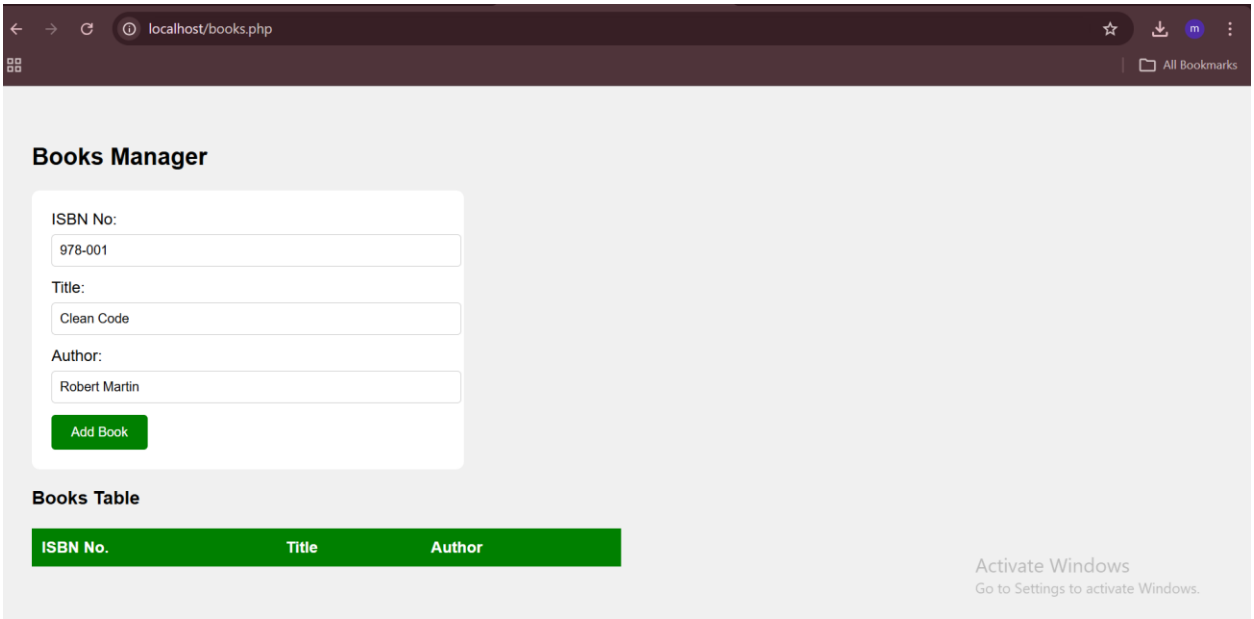


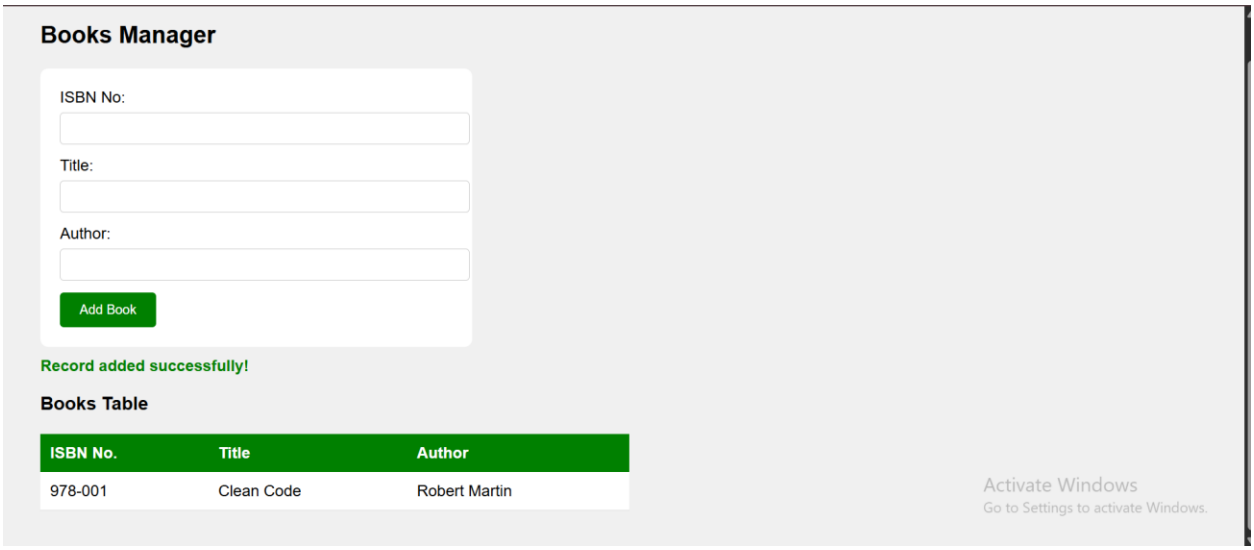
Books.php :

```php
<?php
$conn = mysqli_connect("localhost", "root", "", "lab11");

if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}

$message = "";

if (isset($_POST['submit'])) {
    $isbn   = $_POST['isbn'];
    $title  = $_POST['title'];
    $author = $_POST['author'];

    $sql = "INSERT INTO Books (ISBN_NO, Title, Author)
            VALUES ('$isbn', '$title', '$author')";

    if (mysqli_query($conn, $sql)) {
        $message = "Record added successfully!";
    } else {
        $message = "Error: " . mysqli_error($conn);
    }
```

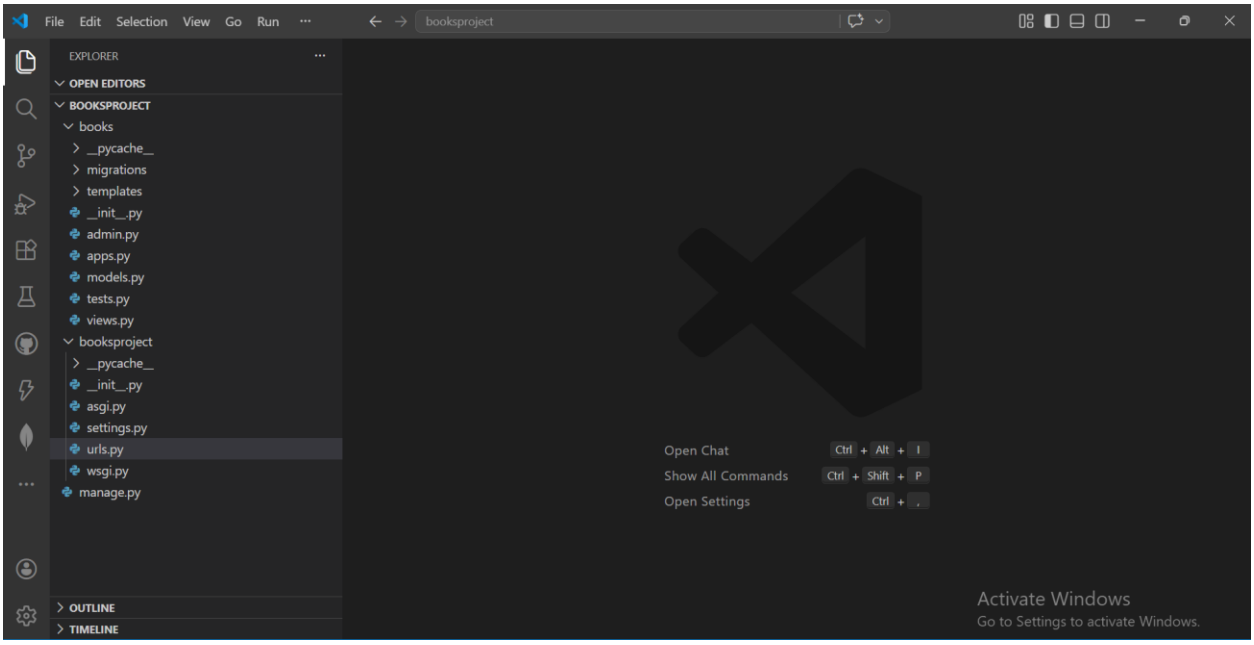
```
}

$result = mysqli_query($conn, "SELECT * FROM Books");
?>

<!DOCTYPE html>
<html>
<head>
    <title>Books - Lab 11</title>
    <style>
        body { font-family: Arial; padding: 30px; background: #f0f0f0; }
        form { background: white; padding: 20px; max-width: 400px; border-radius:
8px; }
        input[type=text] { width: 100%; padding: 8px; margin: 6px 0 12px 0;
border: 1px solid #ccc; border-radius: 4px; }
        input[type=submit] { background: green; color: white; padding: 10px 20px;
border: none; border-radius: 4px; cursor: pointer; }
        table { margin-top: 20px; border-collapse: collapse; background: white;
width: 600px; }
        th { background: green; color: white; padding: 10px; text-align: left; }
        td { padding: 10px; border-bottom: 1px solid #ddd; }
        .msg { color: green; font-weight: bold; margin: 10px 0; }
    </style>
</head>
<body>

<h2>Books Manager</h2>

<form method="POST" action="">
    <label>ISBN No:</label>
    <input type="text" name="isbn"><br>
    <label>Title:</label>
    <input type="text" name="title"><br>
    <label>Author:</label>
    <input type="text" name="author"><br>
    <input type="submit" name="submit" value="Add Book">
</form>

<p class="msg"><?php echo $message; ?></p>

<h3>Books Table</h3>
<table>
    <tr>
        <th>ISBN No.</th>
        <th>Title</th>
```

```
        <th>Author</th>
    </tr>
    <?php while($row = mysqli_fetch_assoc($result)): ?>
    <tr>
        <td><?php echo $row['ISBN_NO']; ?></td>
        <td><?php echo $row['Title']; ?></td>
        <td><?php echo $row['Author']; ?></td>
    </tr>
    <?php endwhile; ?>
</table>

</body>
</html>
```





Task 2:



Models.py :

```
from django.db import models

class Book(models.Model):
    isbn_no = models.CharField(max_length=20, primary_key=True)
    title   = models.CharField(max_length=100)
    author  = models.CharField(max_length=100)
```

```
    class Meta:
        db_table = 'Books'
```

views.py :

```
from django.shortcuts import render, redirect
from .models import Book

def index(request):
    if request.method == 'POST':
        isbn   = request.POST['isbn']
        title  = request.POST['title']
        author = request.POST['author']
        Book.objects.create(isbn_no=isbn, title=title, author=author)
        return redirect('/')

    books = Book.objects.all()
    return render(request, 'books/index.html', {'books': books})
```

urls.py :

```
from django.urls import path
from books import views

urlpatterns = [
    path('', views.index, name='index'),
]
```

Settings.py :

```
"""
Django settings for booksproject project.

Generated by 'django-admin startproject' using Django 6.0.4.

For more information on this file, see
https://docs.djangoproject.com/en/6.0/topics/settings/

For the full list of settings and their values, see
https://docs.djangoproject.com/en/6.0/ref/settings/
"""

from pathlib import Path

BASE_DIR = Path(__file__).resolve().parent.parent
```

```python
SECRET_KEY = 'django-insecure-!@v44l5(#*+q3#mr46@ol=8!xmo4r$9&k*jj#l-b)l+%)y9f($'

DEBUG = True

ALLOWED_HOSTS = []


INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'books',
]

MIDDLEWARE = [
    'django.middleware.security.SecurityMiddleware',
    'django.contrib.sessions.middleware.SessionMiddleware',
    'django.middleware.common.CommonMiddleware',
    'django.middleware.csrf.CsrfViewMiddleware',
    'django.contrib.auth.middleware.AuthenticationMiddleware',
    'django.contrib.messages.middleware.MessageMiddleware',
    'django.middleware.clickjacking.XFrameOptionsMiddleware',
]

ROOT_URLCONF = 'booksproject.urls'

TEMPLATES = [
    {
        'BACKEND': 'django.template.backends.django.DjangoTemplates',
        'DIRS': [],
        'APP_DIRS': True,
        'OPTIONS': {
            'context_processors': [
                'django.template.context_processors.request',
                'django.contrib.auth.context_processors.auth',
                'django.contrib.messages.context_processors.messages',
            ],
        },
    },
]
```

```
WSGI_APPLICATION = 'booksproject.wsgi.application'


DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.mysql',
        'NAME': 'djangodb',
        'USER': 'root',
        'PASSWORD': '',
        'HOST': 'localhost',
        'PORT': '3306',
    }
}


AUTH_PASSWORD_VALIDATORS = [
    {
        'NAME':
'django.contrib.auth.password_validation.UserAttributeSimilarityValidator',
    },
    {
        'NAME': 'django.contrib.auth.password_validation.MinimumLengthValidator',
    },
    {
        'NAME':
'django.contrib.auth.password_validation.CommonPasswordValidator',
    },
    {
        'NAME':
'django.contrib.auth.password_validation.NumericPasswordValidator',
    },
]

LANGUAGE_CODE = 'en-us'

TIME_ZONE = 'UTC'

USE_I18N = True

USE_TZ = True

STATIC_URL = 'static/'
```

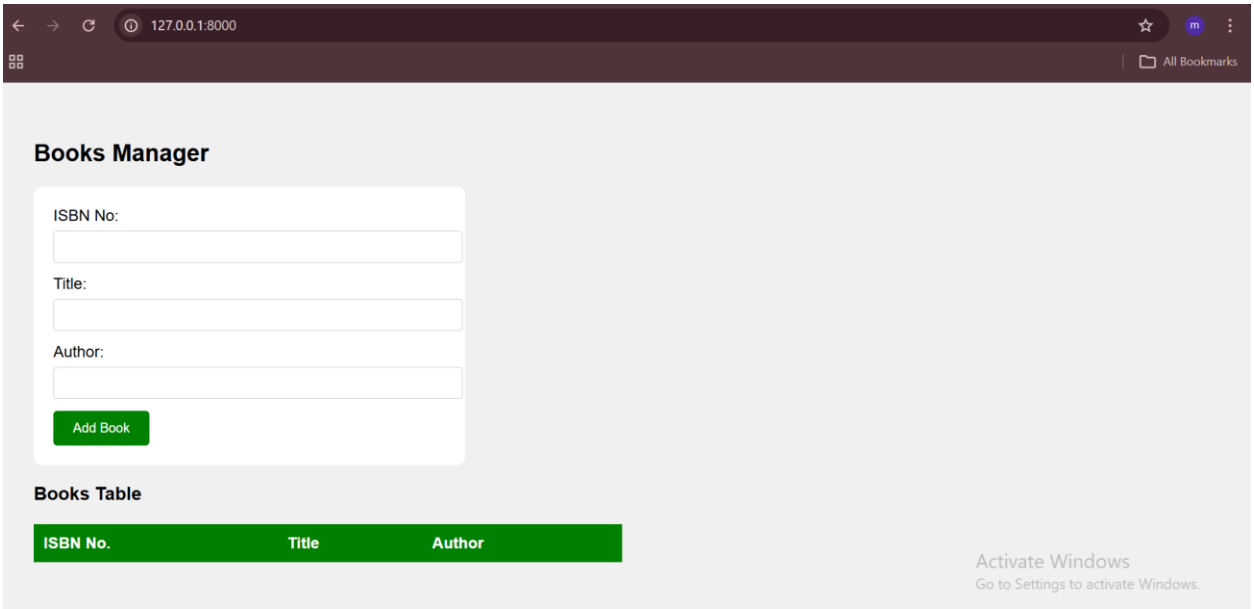
Index.html :

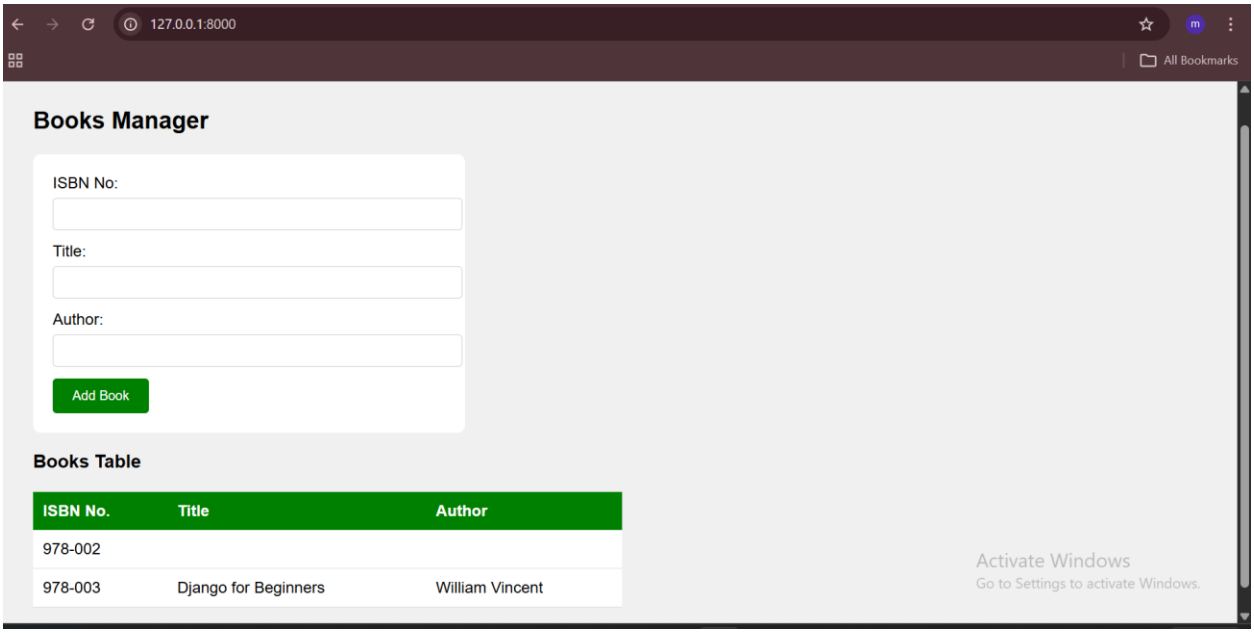
```html
<!DOCTYPE html>
<html>
<head>
    <title>Books - Lab 11 Task 02</title>
    <style>
        body { font-family: Arial; padding: 30px; background: #f0f0f0; }
        form { background: white; padding: 20px; max-width: 400px; border-radius:
8px; }
        input[type=text] { width: 100%; padding: 8px; margin: 6px 0 12px 0;
border: 1px solid #ccc; border-radius: 4px; }
        input[type=submit] { background: green; color: white; padding: 10px 20px;
border: none; border-radius: 4px; cursor: pointer; }
        table { margin-top: 20px; border-collapse: collapse; background: white;
width: 600px; }
        th { background: green; color: white; padding: 10px; text-align: left; }
        td { padding: 10px; border-bottom: 1px solid #ddd; }
    </style>
</head>
<body>

<h2>Books Manager</h2>

<form method="POST" action="">
    {% csrf_token %}
    <label>ISBN No:</label>
    <input type="text" name="isbn"><br>
    <label>Title:</label>
    <input type="text" name="title"><br>
    <label>Author:</label>
    <input type="text" name="author"><br>
    <input type="submit" value="Add Book">
</form>

<h3>Books Table</h3>
<table>
    <tr>
        <th>ISBN No.</th>
        <th>Title</th>
        <th>Author</th>
    </tr>
    {% for book in books %}
    <tr>
        <td>{{ book.isbn_no }}</td>
        <td>{{ book.title }}</td>
        <td>{{ book.author }}</td>
```

```
    </tr>
    {% endfor %}
</table>

</body>
</html>
```

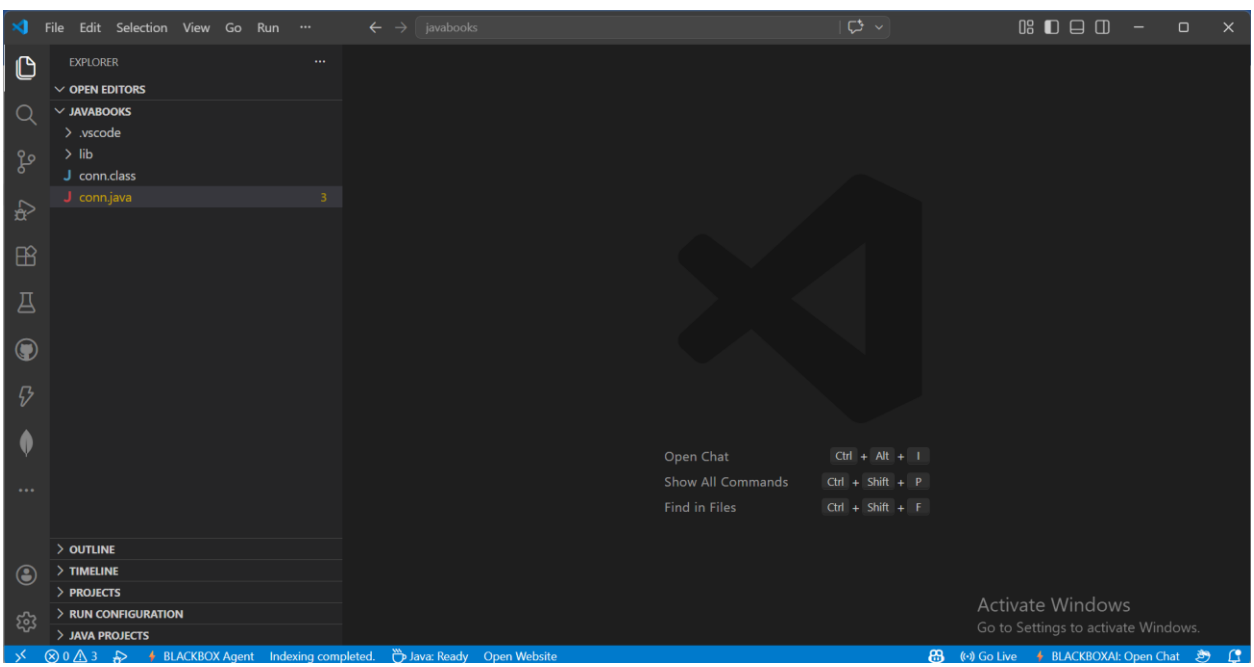
```
Administrator: Command Prompt - python  manage.py runserver
          ^^^^^^^^^^^^^^^^
  File "C:\Python312\Lib\site-packages\django\db\models\manager.py", line 87, in manager_method
    return getattr(self.get_queryset(), name)(*args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Python312\Lib\site-packages\django\db\models\query.py", line 1805, in _insert
    return query.get_compiler(using=using).execute_sql(returning_fields)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Python312\Lib\site-packages\django\db\models\sql\compiler.py", line 1820, in execute_sql
    cursor.execute(sql, params)
  File "C:\Python312\Lib\site-packages\django\db\backends\utils.py", line 102, in execute
    return super().execute(sql, params)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Python312\Lib\site-packages\django\db\backends\utils.py", line 67, in execute
    return self._execute_with_wrappers(
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Python312\Lib\site-packages\django\db\backends\utils.py", line 80, in _execute_with_wrappers
    return executor(sql, params, many, context)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Python312\Lib\site-packages\django\db\backends\utils.py", line 84, in _execute
    with self.db.wrap_database_errors:
  File "C:\Python312\Lib\site-packages\django\db\utils.py", line 91, in __exit__
    raise dj_exc_value.with_traceback(traceback) from exc_value
  File "C:\Python312\Lib\site-packages\django\db\backends\utils.py", line 89, in _execute
    return self.cursor.execute(sql, params)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Python312\Lib\site-packages\django\db\backends\mysql\base.py", line 75, in execute
    return self.cursor.execute(query, args)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Python312\Lib\site-packages\MySQLdb\cursors.py", line 179, in execute
    res = self._query(mogrified_query)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Python312\Lib\site-packages\MySQLdb\cursors.py", line 330, in _query
    db.query(q)
  File "C:\Python312\Lib\site-packages\MySQLdb\connections.py", line 286, in query
    _mysql.connection.query(self, query)
django.db.utils.IntegrityError: (1062, "Duplicate entry '978-002' for key 'PRIMARY'")
[23/Apr/2026 14:06:49] "POST / HTTP/1.1" 500 175972
[23/Apr/2026 14:06:55] "POST / HTTP/1.1" 302 0
[23/Apr/2026 14:06:55] "GET / HTTP/1.1" 200 1563
```

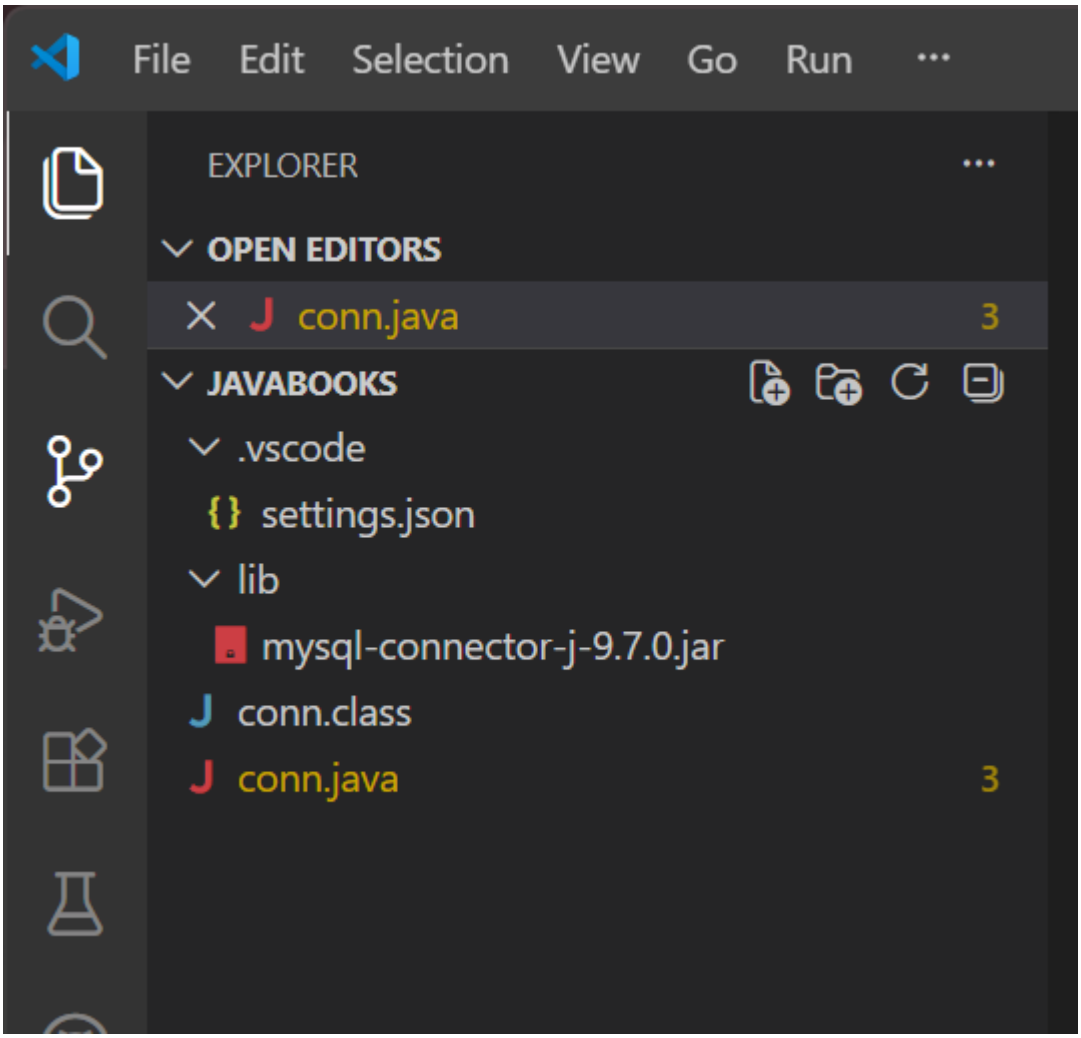
Activate Windows
Go to Settings to activate Windows.




127.0.0.1:8000
All Bookmarks
Books Manager
ISBN No:
Title:
Author:
Add Book
Books Table
ISBN No.
Title
Author
978-002
978-003
Django for Beginners
William Vincent
Activate Windows
Go to Settings to activate Windows.

Task 3:


File
Edit
Selection
View
Go
Run
javabooks
EXPLORER
OPEN EDITORS
JAVABOOKS
.vscode
lib
conn.class
conn.java
OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
JAVA PROJECTS
Open Chat
Ctrl + Alt + I
Show All Commands
Ctrl + Shift + P
Find in Files
Ctrl + Shift + F
Activate Windows
Go to Settings to activate Windows.
BLACKBOX Agent
Indexing completed.
Java: Ready
Open Website
Go Live
BLACKBOXAI: Open Chat



Conn.java :

```
import java.sql.*;

public class conn {
    public static void main(String[] args) {

        String url      = "jdbc:mysql://localhost:3306/lab11";
        String username = "root";
        String password = "";

        Connection con  = null;
        Statement  stmt = null;
        ResultSet  rs   = null;

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            con = DriverManager.getConnection(url, username, password);
            System.out.println("Connected!");

            stmt = con.createStatement();

            String createTable =
                "CREATE TABLE IF NOT EXISTS Books (" +
                "ISBN_NO VARCHAR(20) PRIMARY KEY, " +
                "Title   VARCHAR(100) NOT NULL, " +
                "Author  VARCHAR(100) NOT NULL)";
            stmt.execute(createTable);
            System.out.println("Table ready!");

            String insert1 = "INSERT IGNORE INTO Books VALUES ('978-001', 'Clean
Code', 'Robert Martin')";
            String insert2 = "INSERT IGNORE INTO Books VALUES ('978-002', 'Java
Programming', 'Herbert Schildt')";
            String insert3 = "INSERT IGNORE INTO Books VALUES ('978-003', 'Django
for Beginners', 'William Vincent')";

            stmt.execute(insert1);
            stmt.execute(insert2);
            stmt.execute(insert3);
            System.out.println("Records inserted successfully!");

            rs = stmt.executeQuery("SELECT * FROM Books");
            System.out.println("\nShowing data!");
            System.out.println("-----------------------------------------------
-");
            System.out.printf("%-22s %-25s %-20s%n", "ISBN_NO", "Title",
"Author");
            System.out.println("-----------------------------------------------
-");

            while (rs.next()) {
                System.out.printf("%-22s %-25s %-20s%n",
                    rs.getString("ISBN_NO"),
                    rs.getString("Title"),
                    rs.getString("Author"));
            }
            System.out.println("-----------------------------------------------
-");

        } catch (Exception ex) {
            System.out.println("Error: " + ex.getMessage());
```

```java
            ex.printStackTrace();

        } finally {
            try {
                if (rs   != null) rs.close();
                if (stmt != null) stmt.close();
                if (con  != null) con.close();
                System.out.println("\nConnection closed.");
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
}
```

```
PROBLEMS 3   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   MYBATISX                    Run: conn

PS C:\Users\PMLS\Downloads\New folder\javabooks>  & 'C:\Program Files\Java\jdk-24\bin\java.exe' '@C:\Users\PMLS\AppData\Local\Temp\cp_2un2e1s7joljgtvg51csvw3r9.argfile' 'conn'
Connected!
Table ready!
Records inserted successfully!

Showing data!
--------------------------------------------------
ISBN_NO              Title                   Author
--------------------------------------------------
978-001              Clean Code              Robert Martin
978-002              Java Programming        Herbert Schildt
978-003              Django for Beginners    William Vincent
--------------------------------------------------

Connection closed.
PS C:\Users\PMLS\Downloads\New folder\javabooks>
```